

Square matrices that square to zero: a proof of Branman’s conjecture on OEIS A087726

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Abstract

Let $a(n)$ be the number of $X \in M_2(\mathbb{Z}/n\mathbb{Z})$ with $X^2 \equiv 0$, OEIS A087726. In March 2013 B. Branman conjectured that $a(n) = n^2$ if and only if n is squarefree; E. M. Schmidt confirmed the “if” direction the following day, and the converse has been open since. We prove it. The argument needs no formula for $a(p^k)$: since a is multiplicative and $a(p) = p^2$, it suffices to show $a(p^k) > p^{2k}$ for $k \geq 2$, and for that a lower bound is enough. Every trace-zero matrix $X = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}$ satisfies $X^2 = (a^2 + bc)I$, so $a(p^k) \geq N(k) := \#\{(a, b, c) \bmod p^k : a^2 + bc \equiv 0\}$, and N obeys the two-step recursion $N(k) = (p^2 - 1)p^{2k-2} + p^3N(k-2)$, from which $N(k) > p^{2k}$ for $k \geq 2$ falls out in two lines.

1 The conjecture

OEIS A087726 (Yuval Dekel, September 2003) is

$$a(n) = \#\{X \in M_2(\mathbb{Z}/n\mathbb{Z}) : X^2 \equiv 0 \bmod n\}, \quad a(1), \dots, a(9) = 1, 4, 9, 28, 25, 36, 49, 112, 153.$$

The entry carries

Conjecture: $a(n) = n^2$ if and only if n is squarefree. — Ben Branman, Mar 22 2013

Preceding conjecture is true in the case where n is squarefree. — Eric M. Schmidt, Mar 23 2013

and, separately, an unproven closed form for $a(p^k)$ observed by R. Israel in June 2015, of which only the case $k = 1$ has a recorded proof. As of the “Last modified” line on the live entry (23 August 2026) Branman’s conjecture is still recorded as unproven. We prove it below without using Israel’s formula.

2 Reduction

Lemma 1 (multiplicativity). *If $\gcd(m, n) = 1$ then $a(mn) = a(m)a(n)$.*

Proof. The Chinese remainder theorem gives a ring isomorphism $\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$, hence $M_2(\mathbb{Z}/mn\mathbb{Z}) \cong M_2(\mathbb{Z}/m\mathbb{Z}) \times M_2(\mathbb{Z}/n\mathbb{Z})$, and $X^2 = 0$ holds if and only if it holds in each factor. $\square$

Lemma 2 (the trace-zero family). *For any commutative ring R and $a, b, c \in R$, the matrix $X = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}$ satisfies $X^2 = (a^2 + bc)I$. Hence $X^2 = 0$ if and only if $a^2 + bc = 0$.*

Proof. Direct multiplication: the diagonal entries are $a^2 + bc$ and $cb + a^2$, and the off-diagonal entries are $ab - ba = 0$ and $ca - ac = 0$. $\square$

For a prime p and $k \geq 0$ set

$$N(k) = \#\{(a, b, c) \in (\mathbb{Z}/p^k\mathbb{Z})^3 : a^2 + bc \equiv 0 \pmod{p^k}\},$$

with the convention $N(0) = 1$. Lemma 2 gives immediately

$$a(p^k) \geq N(k). \quad (1)$$

Lemma 3 (the prime case). $a(p) = N(1) = p^2$.

Proof. Over the field $\mathbb{F}_p$, if $X^2 = 0$ and $X \neq 0$ then the minimal polynomial of X is t^2 , so the characteristic polynomial is t^2 and $\text{tr } X = \det X = 0$; and $X = 0$ satisfies this too. Conversely if $\text{tr } X = \det X = 0$ then $X^2 = 0$ by Cayley–Hamilton. So the solutions are exactly the trace-zero matrices with $a^2 + bc = 0$, i.e. $a(p) = N(1)$. To count $N(1)$: for each of the $p^2 - p$ pairs (a, b) with $b \neq 0$ there is one c ; for $b = 0$ the condition is $a^2 = 0$, i.e. $a = 0$, and c is free, giving p more. Total $p^2 - p + p = p^2$. $\square$

Proposition 4. *Suppose $a(p^k) > p^{2k}$ for every prime p and every $k \geq 2$. Then Branman’s conjecture holds.*

Proof. Write $n = \prod p^{k_p}$. By Lemma 1, $a(n) = \prod_p a(p^{k_p})$, while $n^2 = \prod_p p^{2k_p}$. By Lemma 3 the factors agree when $k_p = 1$, and by hypothesis $a(p^{k_p}) > p^{2k_p}$ when $k_p \geq 2$. All factors are positive, so $a(n) \geq n^2$ with equality precisely when no k_p exceeds 1, i.e. precisely when n is squarefree. $\square$

3 The counting recursion

Lemma 5. *For every prime p and every $k \geq 2$,*

$$N(k) = (p^2 - 1)p^{2k-2} + p^3 N(k-2),$$

with $N(0) = 1$ and $N(1) = p^2$.

Proof. Split the solutions according to whether p divides all of a, b, c .

Imprimitive solutions. If $a \equiv b \equiv c \equiv 0 \pmod{p}$, write $(a, b, c) = (pa', pb', pc')$ with a', b', c' ranging over $\mathbb{Z}/p^{k-1}\mathbb{Z}$; the condition becomes $p^2(a'^2 + b'c') \equiv 0 \pmod{p^k}$, i.e. $a'^2 + b'c' \equiv 0 \pmod{p^{k-2}}$. That condition depends on (a', b', c') only modulo p^{k-2} , and each residue modulo p^{k-2} has p lifts modulo p^{k-1} in each coordinate. So the imprimitive solutions number $p^3 N(k-2)$.

Primitive solutions. Let $Q(a, b, c) = a^2 + bc$, with gradient $\nabla Q = (2a, c, b)$. If (a, b, c) is a solution modulo p with $\nabla Q \equiv 0$ then $b \equiv c \equiv 0$; for odd p also $a \equiv 0$, and for $p = 2$ the congruence $Q \equiv 0$ forces $a^2 \equiv 0$, hence $a \equiv 0$ as well. So every *primitive* solution modulo p is a nonsingular point of Q , and by Hensel’s lemma each lifts to exactly $p^{2(k-1)}$ solutions modulo p^k . By Lemma 3 there are $N(1) = p^2$ solutions modulo p , of which one — the origin — is imprimitive, so $p^2 - 1$ are primitive. Hence the primitive solutions modulo p^k number $(p^2 - 1)p^{2k-2}$. $\square$

Lemma 6. $N(k) \geq p^{2k}$ for every $k \geq 0$, and $N(k) > p^{2k}$ for every $k \geq 2$.

Proof. Rewrite Lemma 5 as

$$N(k) - p^{2k} = p^3 N(k-2) - p^{2k-2} \quad (k \geq 2), \quad (2)$$

using $(p^2 - 1)p^{2k-2} = p^{2k} - p^{2k-2}$.

First, $N(k) \geq p^{2k}$ for all $k \geq 0$, by induction in steps of two. The bases are $N(0) = 1 = p^0$ and $N(1) = p^2$. For $k \geq 2$, the inductive hypothesis $N(k-2) \geq p^{2k-4}$ and (2) give

$$N(k) - p^{2k} \geq p^3 p^{2k-4} - p^{2k-2} = p^{2k-1} - p^{2k-2} = p^{2k-2}(p-1) > 0.$$

This is a strict inequality, which simultaneously proves the second claim for $k \geq 2$. $\square$

4 The theorem

Theorem 7. For every $n \geq 1$, $a(n) \geq n^2$, with equality if and only if n is squarefree.

Proof. By (1) and Lemma 6, $a(p^k) \geq N(k) > p^{2k}$ for every $k \geq 2$. Proposition 4 then gives the statement. $\square$

Remark (what is and is not proved). Only a lower bound on $a(p^k)$ is used, so this argument says nothing about the exact value of $a(p^k)$ and in particular does not prove Israel's observed formula, which remains open. The gap is real: for instance $a(4) = 28$ while $N(2) = 20$, the difference being the solutions with nonzero trace, which exist over $\mathbb{Z}/p^k\mathbb{Z}$ for $k \geq 2$ although not over a field. The conjecture simply does not need them.

Remark (honest assessment). The three ingredients — Chinese remainder, Cayley–Hamilton in the form $\begin{pmatrix} a & b \\ c & -a \end{pmatrix}^2 = (a^2 + bc)I$, and Hensel lifting for the ternary form $a^2 + bc$ — are all standard, and the write-up is short. The reason the statement survived thirteen years is presumably that the natural attack is Israel's exact formula for $a(p^k)$, which is genuinely harder than the conjecture itself. The appropriate destination is an OEIS comment.

5 Verification

A verification script, written separately from this note, checks every claim and prints every number quoted. It (i) recomputes $a(1), \dots, a(12)$ by exhaustive search over $M_2(\mathbb{Z}/n\mathbb{Z})$ and matches the published DATA exactly; (ii) confirms Lemma 1 on all coprime pairs with $mn \leq 40$, with 0 mismatches; (iii) confirms $a(p) = p^2$ for $p = 2, 3, 5, 7$; (iv) confirms the recursion of Lemma 5 against a brute-force count of $N(k)$ for all $p^k \leq 27$, with 0 mismatches, giving for $p = 2$ the values $N(0..8) = 1, 4, 20, 80, 352, 1408, 5888, 23552, 96256$ and for $p = 3$ $N(0..6) = 1, 9, 99, 891, 8505, 76545, 702027$; (v) confirms $N(k) \geq p^{2k}$ for $0 \leq k \leq 24$ and $p \in \{2, 3, 5, 7, 11, 13, 101, 1009\}$, strict for $k \geq 2$, together with identity (2); (vi) checks $a(p^k) \geq N(k) > p^{2k}$ directly for $p^k \in \{4, 8, 9, 16, 25, 27\}$, e.g. $a(16) = 640 \geq 352 > 256$; and (vii) confirms Theorem 7 itself — that $a(n) = n^2$ exactly for the squarefree $n \leq 40$ — with 0 mismatches.

References

- [1] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A087726>, entry A087726, consulted 24 August 2026.
- [2] J.-P. Serre, *A Course in Arithmetic*, Springer, 1973 (for Hensel's lemma and counts of quadratic-form solutions).

A Suggested OEIS comment (plain text, house style)

Branman's conjecture is true. Since a is multiplicative (CRT) and $a(p) = p^2$ (over a field, $X^2 = 0$ iff $\text{tr } X = \det X = 0$), it suffices to prove $a(p^k) > p^{(2k)}$ for $k \geq 2$, and for that a lower bound suffices.

Every trace-zero matrix $\begin{bmatrix} a & b \\ c & -a \end{bmatrix}$ satisfies $\begin{bmatrix} a & b \\ c & -a \end{bmatrix}^2 = (a^2 + bc)I$, so $a(p^k) \geq N(k) := \#\{(a, b, c) \bmod p^k : a^2 + bc = 0 \bmod p^k\}$.

N satisfies $N(0) = 1$, $N(1) = p^2$, and for $k \geq 2$

$$N(k) = (p^2 - 1)p^{(2k-2)} + p^3N(k-2).$$

(Split by whether p divides all of a, b, c . Imprimitve triples are p times a solution mod $p^{(k-2)}$, contributing $p^3N(k-2)$. Every primitive solution mod p is a nonsingular point of $a^2 + bc$ - the gradient $(2a, c, b)$ vanishing forces $b = c = 0$ and then $a = 0$, for $p = 2$ as well - so Hensel lifts each of the $p^2 - 1$ primitive solutions mod p to $p^{(2k-2)}$ solutions mod p^k .)

Rewriting, $N(k) - p^{(2k)} = p^3N(k-2) - p^{(2k-2)}$. Induction in steps of two from $N(0) = 1$ and $N(1) = p^2$ gives $N(k) \geq p^{(2k)}$ for all k , and then $N(k) - p^{(2k)} \geq p^{(2k-1)} - p^{(2k-2)} > 0$ for $k \geq 2$. Hence $a(p^k) > p^{(2k)}$ for $k \geq 2$, and $a(n) = n^2$ exactly for squarefree n .

This does not prove Israel's formula for $a(p^k)$; it only bounds $a(p^k)$ below. For example $a(4) = 28$ while $N(2) = 20$, the difference being the nonzero-trace solutions, which the conjecture does not need.